

Gain equations

- The power gain is:

$$G = \frac{P_L}{P_{in}} = \frac{|S_{21}|^2(1-|\Gamma_L|^2)}{(1-|\Gamma_{in}|^2)|1-S_{22}\Gamma_L|^2}$$

- The available gain is:

$$G_A = \frac{P_{AVN}}{P_{AVS}} = \frac{|S_{21}|^2(1-|\Gamma_s|^2)}{|1-S_{11}\Gamma_s|^2(1-|\Gamma_{out}|^2)}$$

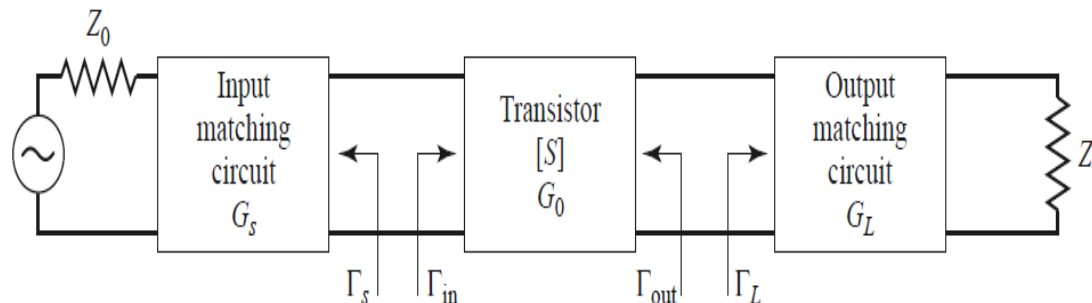
Low noise amplifier design

- The transducer power gain is:

$$G_T = \frac{P_L}{P_{AVS}} = \frac{(1-|\Gamma_s|^2)}{|1-\Gamma_{in}\Gamma_s|^2} |S_{21}|^2 \frac{(1-|\Gamma_L|^2)}{|1-S_{22}\Gamma_L|^2}$$

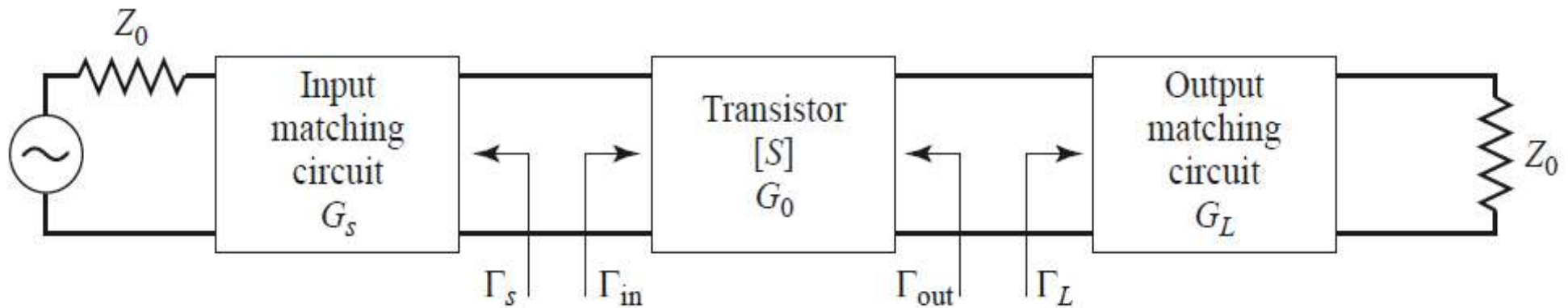
$$G_T = \frac{P_L}{P_{AVS}} = \frac{(1-|\Gamma_s|^2)}{|1-s_{11}\Gamma_s|^2} |S_{21}|^2 \frac{(1-|\Gamma_L|^2)}{|1-\Gamma_{out}\Gamma_L|^2}$$

**Maximum gain
amplifier design**





Maximum gain design (conjugate matching)



- $\Gamma_{in} = \Gamma_s^*$ and $\Gamma_{out} = \Gamma_L^*$

- $\Gamma_s^* = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1-S_{22}\Gamma_L}$

- $\Gamma_L^* = S_{22} + \frac{S_{12}S_{21}\Gamma_s}{1-S_{11}\Gamma_s}$

- $\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$

- $\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$

$$G_{Tmax} = G_{smax} G_o G_{Lmax}$$

$$G_{Tmax} = \frac{1}{(1 - |\Gamma_s|^2)} |S_{21}|^2 \frac{(1 - |\Gamma_L|^2)}{|1 - S_{22}\Gamma_L|^2}$$

$$G_{Tmax} = \frac{|S_{21}|}{|S_{12}|} (K - \sqrt{K^2 - 1})$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$

$$C_2 = S_{22} - \Delta S_{11}^*$$